DEPARTMENT OF TECHNOLOGY

Machine Learning

Polynomial Regression

Learning Objective

After this lecture, students will be able to:

- Understand why Polynomial Regression is required.
- Understand the basic idea of polynomial features.
- Write the equation of a Polynomial Regression model.
- Understand the meaning of polynomial degree.
- Understand the relationship between degree and model complexity.
- Understand the basic idea of underfitting and overfitting.

1 Why Polynomial Regression?

In Linear Regression, the model learns a linear relationship between the input and output.

However, in some datasets, the relationship between the input feature and target variable may not be approximately linear.

For example, consider a situation where the output first increases slowly and then increases rapidly.

A straight-line relationship may not represent such a pattern well.

Therefore, we need a model that can represent a **curved relationship**.

This leads to:

Polynomial Regression

2 Basic Idea

Polynomial Regression extends the regression model by introducing powers of the input feature.

For one input feature x, a degree-2 polynomial model is:

$$\hat{y} = w_0 + w_1 x + w_2 x^2 \tag{1}$$

Here, the additional term is:

$$x^2$$

This allows the model to represent a curved relationship between x and $\hat{y}$.

3 Polynomial Regression Equation

The general Polynomial Regression equation is:

$$\boxed{\hat{y} = w_0 + w_1 x + w_2 x^2 + \cdots + w_n x^n} \tag{2}$$

where:

- x = input feature
- $\hat{y}$ = predicted output
- w_0 = intercept
- $w_1, w_2, \ldots, w_n$ = model coefficients
- n = degree of the polynomial

4 What is Polynomial Degree?

The **degree** of a polynomial is the highest power of the input feature present in the model.

Degree 1

$$\hat{y} = w_0 + w_1 x \tag{3}$$

This corresponds to a linear model.

Degree 2

$$\boxed{\hat{y} = w_0 + w_1 x + w_2 x^2} \tag{4}$$

This is a quadratic polynomial.

Degree 3

$$\boxed{\hat{y} = w_0 + w_1 x + w_2 x^2 + w_3 x^3} \tag{5}$$

This is a cubic polynomial.
Therefore:

$$\boxed{\text{Higher Degree} \rightarrow \text{More Flexible Model}} \tag{6}$$

5 Simple Example

Consider the following dataset:

x	y
1	3
2	7
3	13
4	21
5	31

Observe the values of y as x increases.
The change in y is not constant.

Important

Classroom Activity
Observe the dataset and answer:

1. Calculate the difference between consecutive values of y.
2. Is the increase in y constant?
3. Does the relationship appear to be linear?
4. Which polynomial degree might be suitable for representing this pattern?

Do not calculate the model coefficients.
The objective is only to identify the type of relationship.

6 Polynomial Features

A **polynomial feature** is a new feature created from an existing numerical feature by applying a power to it.

Suppose the original feature is:

$$x$$

For a degree-2 polynomial model, we create an additional feature by squaring each value of x:

$$\boxed{x^2}$$

We **do not replace** the original feature x. We keep x and add x^2 as another feature.

6.1 Example

Suppose the original feature contains the following values:

Original Feature: x
1
2
3
4

For a degree-2 polynomial model, calculate a new feature:

$$x^2$$

Therefore:

Original Feature x	**Polynomial Feature** x^2
1	$1^2 = 1$
2	$2^2 = 4$
3	$3^2 = 9$
4	$4^2 = 16$

6.2 New Feature Matrix

Originally, the model has only one feature:

$$X = [x]$$

After creating the polynomial feature:

$$\boxed{X_{\text{new}} = [x,\ x^2]}$$

For the above example:

$$\boxed{X_{\text{new}} = \begin{bmatrix} 1 & 1 \\ 2 & 4 \\ 3 & 9 \\ 4 & 16 \end{bmatrix}}$$

Therefore, the model now receives two input features:

$$\boxed{x \qquad \text{and} \qquad x^2}$$

6.3 Degree 3

If we choose degree 3, we create:

$$x, \quad x^2, \quad x^3$$

Therefore:

$$\boxed{X_{\text{new}} = [x,\ x^2,\ x^3]}$$

The original feature x is retained, and the new polynomial features are added.
For example:

x	x^2	x^3
1	1	1
2	4	8
3	9	27
4	16	64

6.4 How Polynomial Features Change the Model

Once the polynomial features are created, they are used as inputs to the regression model.

For degree 2:

$$\boxed{\hat{y} = w_0 + w_1 x + w_2 x^2} \tag{7}$$

For degree 3:

$$\boxed{\hat{y} = w_0 + w_1 x + w_2 x^2 + w_3 x^3} \tag{8}$$

Thus, Polynomial Regression uses the original feature along with its polynomial features.

6.5 Important Idea

Important

Polynomial features are additional features created from existing numerical features.

For example:

$$x \rightarrow [x,\ x^2]$$

for degree 2, and

$$x \rightarrow [x,\ x^2,\ x^3]$$

for degree 3.

The original feature is not removed.

7 Polynomial Degree and Model Complexity

The degree controls how flexible the model can be.

For example:

$$\text{Degree } 1 \rightarrow \text{Low Complexity} \tag{9}$$

$$\text{Degree } 2 \rightarrow \text{Higher Complexity} \tag{10}$$

$$\text{Degree } 3 \rightarrow \text{Even Higher Complexity} \tag{11}$$

Therefore:

$$\boxed{\text{Polynomial Degree} \rightarrow \text{Model Complexity}} \tag{12}$$

8 Too Simple vs Too Complex

Choosing the polynomial degree is important.

If the degree is too low, the model may not be flexible enough to capture the underlying pattern.

This is called:

$$\boxed{\text{Underfitting}} \tag{13}$$

If the degree is too high, the model may become excessively complex and start fitting noise in the training data.

This is called:

$$\boxed{\text{Overfitting}} \tag{14}$$

9 Underfitting

Underfitting occurs when the model is too simple to capture the important pattern in the data.

$$\boxed{\text{Model Too Simple} \rightarrow \text{Underfitting}} \tag{15}$$

Characteristics:

- Model has insufficient complexity.
- Important patterns are missed.
- Performance can be poor even on training data.

10 Overfitting

Overfitting occurs when the model is too complex and learns the training data too closely.

The model may learn:

$$\text{Pattern} + \text{Noise}$$

instead of learning only the general pattern.

$$\boxed{\text{Model Too Complex} \rightarrow \text{Overfitting}} \tag{16}$$

Characteristics:

- Very flexible model.
- Very good fit to training data.
- Poor performance on unseen data.

11 Generalization

A good Machine Learning model should not simply memorize the training data.

It should learn the underlying pattern and perform well on new, unseen observations.

This ability is called:

$$\boxed{\text{Generalization}} \tag{17}$$

Therefore, our goal is:

$$\boxed{\text{Learn the Pattern} \rightarrow \text{Generalize to New Data}} \tag{18}$$

12 Choosing the Polynomial Degree

The goal is not to choose the highest possible degree.

The goal is to choose an appropriate degree that captures the important pattern without making the model unnecessarily complex.

Conceptually:

Too Simple	$\rightarrow$	Underfitting
$\downarrow$		$\downarrow$
Appropriate Complexity	$\rightarrow$	Good Generalization
$\downarrow$		$\downarrow$
Too Complex	$\rightarrow$	Overfitting

13 Key Terms

Term	**Meaning**
Polynomial Regression	Regression model that uses polynomial terms of the input features.
Degree	Highest power of the input feature in the model.
Polynomial Feature	A transformed feature such as x^2 or x^3.
Model Complexity	The flexibility of a model to represent patterns in data.
Underfitting	The model is too simple to capture the important pattern.
Overfitting	The model is too complex and fits the training data too closely.
Generalization	Ability of the model to perform well on unseen data.

14 Important Points

Important

1. Polynomial Regression is used when a simple linear relationship is not sufficient.
2. Polynomial features are created using powers of the original feature.
3. The degree determines the complexity of the polynomial model.
4. Increasing the degree increases model flexibility.
5. Very low complexity can lead to underfitting.
6. Very high complexity can lead to overfitting.
7. The objective is to achieve good generalization.

15 Think and Discuss

1. Why do we need Polynomial Regression?
2. What is a polynomial feature?
3. What does the degree of a polynomial represent?
4. What happens to model complexity when the polynomial degree increases?
5. What is underfitting?
6. What is overfitting?
7. Why is a very high-degree polynomial not necessarily a better model?
8. What do we mean by generalization?

16 Connection to the Next Topic

We have seen that increasing polynomial degree increases model complexity.
However, a highly complex model can overfit the training data.
This raises the next question:

Important

Can we control model complexity and reduce overfitting?

This leads to:

Regularization

The next topic will introduce:

Ridge Regression and Lasso Regression

and show how a penalty term can be added to the cost function to control model complexity.